

We *Built*
an Agent

This gap is the **most expensive gap** in enterprise AI today.

Harness Engineering

We *Run* an
Agent in
Production

The Reality

Enterprises are deploying agents. Almost none are scaling them.

97% → 12%

of executives report deploying AI agents this year.
Only 12% reach production at scale

78% → 14%

of enterprises have agent pilots running. Only 14% have hit
production scale.

80%

of enterprise applications now embed at least one AI agent but
most still can't operate them reliably.

The gap between "**we built an agent**" and "**we run an agent in production**" is the **most expensive gap** in enterprise AI today.

What Breaks

A great demo doesn't make a production system.

Enterprise agents fail in production for reasons that have little to do with model quality:

Context drift: agents lose the thread on long tasks

Silent failures: wrong answers that look right. Datadog reports ~1 in 20 production AI requests already fail this way

Unbounded tool use: autonomy without guardrails becomes a liability

Doom loops: agents confidently retrying the same broken approach 10+ times

No observability: failures compound invisibly across sessions

A better model on the same broken scaffolding produces the same broken outcomes.

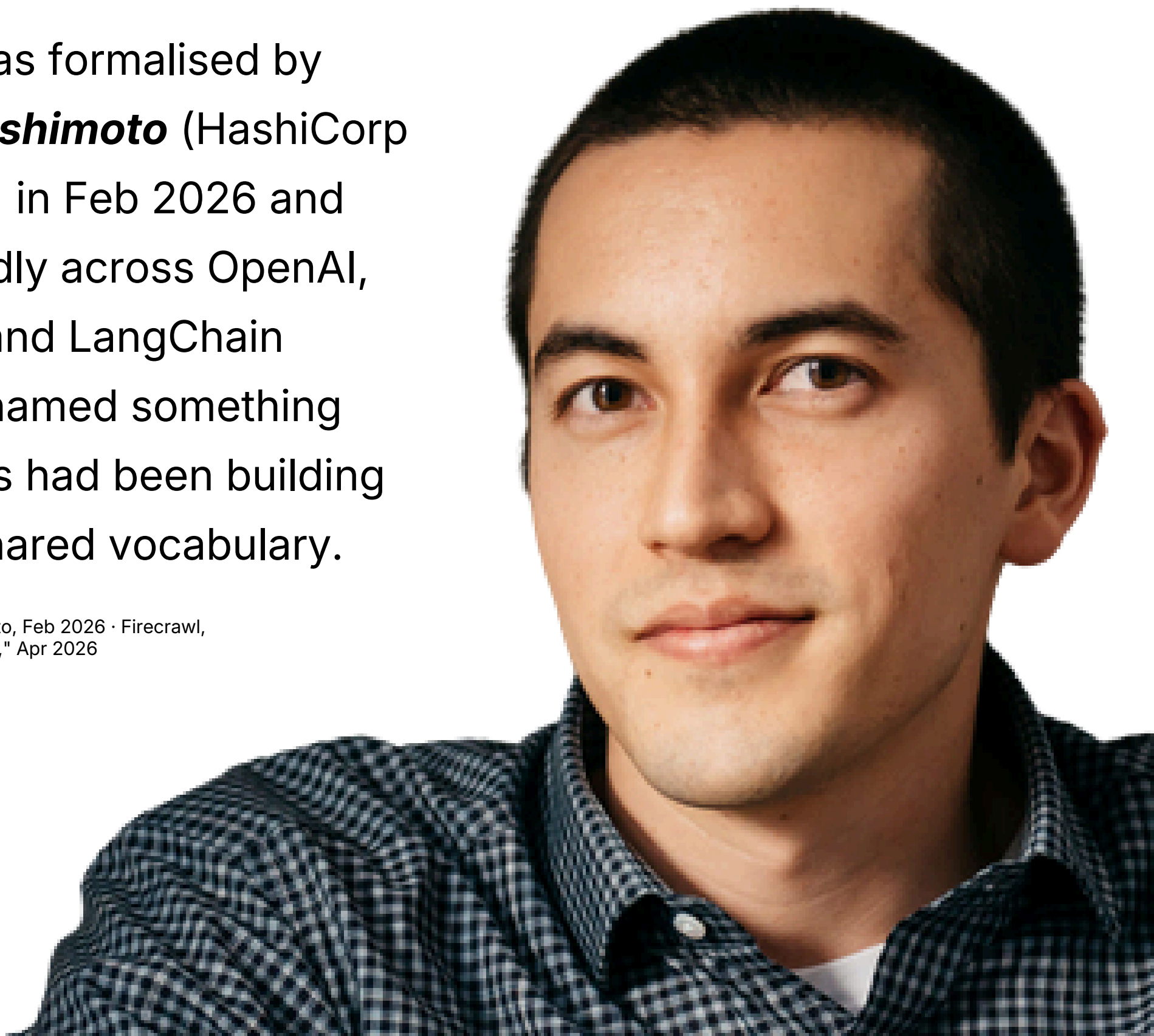
The Concept

What is **Harness Engineering**?

The discipline of designing the systems, constraints, and feedback loops that wrap around an AI agent to make it reliable in production. The model reasons. The harness is everything else: tools, memory, context management, guardrails, evaluation, observability

The term was formalised by ***Mitchell Hashimoto*** (HashiCorp co-founder) in Feb 2026 and spread rapidly across OpenAI, Anthropic, and LangChain because it named something practitioners had been building without a shared vocabulary.

Sources: Mitchell Hashimoto, Feb 2026 · Firecrawl, "What Is an Agent Harness," Apr 2026



A harness is not one thing. It's **six**.

Memory & Context

What does the agent see right now? Compression, retrieval, state stores.

Tools & Skills

What can the agent actually do? APIs, file systems, reusable methods.

Orchestration

Who plans, who executes, who hands off. The workflow choreography.

Infra & Guardrails

Sandboxing, permissions, rollback, safety rails. The runtime boundary.

Evaluation

Self-checks, evaluator agents, confidence scoring, human-in-the-loop gates

Observability

Tracing, session replay, drift detection - failures become fixable, not invisible.

The Shift

From model-centric to infrastructure-centric.

2023

Prompt Engineering

"How do I phrase this question?"

2025

Context Engineering

"What information should the model see?"

2026

Harness Engineering

"How do I build the system that makes the model reliably useful?"

The harness is the dataset. Competitive advantage is now the trajectories your harness captures.

- Philipp Schmid, AI Engineer Lead, Google DeepMind

The Pattern

The teams that scaled agents share one pattern.

A March 2026 survey of 650 enterprise tech leaders found successful scalers didn't spend more on AI overall. They spent differently:

- ✓ **More** on evaluation infrastructure
- ✓ **More** on monitoring & tracing tools
- ✓ **More** on operational staffing (dedicated AI ops)
- ✗ **Less** on model selection and prompt tweaking

Forrester Root Cause Analysis

Negative-ROI agent deployments: 41% failed due to unclear success criteria, 33% to insufficient tool/data access, 26% to evaluation drift. ***None are model-quality problems. All are harness problems.***

If you're building enterprise agents in 2026, start here.

Define the failure modes before the success criteria

What does "wrong" look like in your domain? Specify before you ship.

Build an AGENTS.md

A single file at the root of your project that tells every agent how to behave conventions, anti-patterns, escalation rules. Grow it one rule at a time.

Instrument before you scale

Tracing, structured logs, session replay. You can't govern what you can't see.

Add confidence scoring + human-in-the-loop gates

Route low-confidence outputs to review. Define thresholds by task risk.

Treat every agent failure as a system problem

Fix the harness once, not the prompt fifty times.

Plan for model upgrades

After a model release, the first question isn't "what can I add?" - it's "what can I remove?"

The next phase of enterprise AI is being decided **at the harness layer.**

The teams that win this phase won't be the ones with the best model. They'll be the ones with the most disciplined infrastructure around the model.

Harness engineering is how that discipline gets built.

To know more, checkout DHS **Hack Session** on:

Harness Engineering for Enterprise AI Agents



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The Leela Bhartiya City Bengaluru, India

Abhishek Kumar

Senior Director Data Science

publicis
sapient

